

Instruction Prompt for Claude Code — CDC 2026 Synthetic Experiment

Overview

You are implementing **Algorithm 1** from the attached CDC 2026 draft paper: "*Online Distributed Resource Allocation with Persistent Safety Guarantees*."

Read the entire paper first — especially Sections III–V (Problem Formulation, Algorithm, Theory) and the Appendix proofs — before writing any code. The parameter choices below are derived directly from the analysis; you must verify they satisfy all stated conditions before running.

1. Synthetic Problem Definition

We use a **separable quadratic objective with a linear aggregate constraint** over m agents. Each agent i controls a scalar decision variable $x_i \in X_i := [0, 1]$.

Local cost (objective):

$$f_i(x_i) = (1/2) * \alpha_i * (x_i - a_i)^2$$

- Gradient: $\nabla f_i(x_i) = \alpha_i * (x_i - a_i)$
- Each f_i is α_i -strongly convex and $L_{\{f_i\}}$ -Lipschitz on $[0, 1]$.

Local resource usage (constraint):

$$g_i(x_i) = e_i * x_i - b_i$$

- Gradient: $\nabla g_i(x_i) = e_i$ (constant)
- g_i is convex and e_i -Lipschitz.

Aggregate constraint:

$$G(x) = \sum_i g_i(x_i) = \sum_i (e_i * x_i - b_i) \leq 0$$

Team objective:

$$F(x) = \sum_i f_i(x_i)$$

Design intent:

Parameters are chosen so the **unconstrained** minimisers $x_i = a_i$ **violate** the budget ($G(a) > 0$), forcing the constraint to be active at the true optimum x^* . We also choose **large α_i and small e_i** so that the ratio M/L_G is large, which allows more local steps T per round (Condition (13)). This is the natural regime for the paper: robots have strong local preferences but individually modest resource consumption.

2. Parameters

Use $m = 2$ agents.

Agent	α_i	a_i	e_i	b_i
1	5.0	0.9	0.3	0.1
2	4.0	0.85	0.4	0.1

Verify constraint is violated at unconstrained optimum:

$$\begin{aligned} G(a) &= (0.3 * 0.9 - 0.1) + (0.4 * 0.85 - 0.1) \\ &= 0.17 + 0.24 = 0.41 > 0 \quad \checkmark \end{aligned}$$

Compute x^* via KKT:

The Lagrangian is $L = \sum_i (1/2) * \alpha_i * (x_i - a_i)^2 + \lambda * \sum_i (e_i * x_i - b_i)$.

KKT stationarity: $\alpha_i * (x_i - a_i) + \lambda * e_i = 0 \Rightarrow x_i^* = a_i - \lambda * e_i / \alpha_i$

Complementary slackness: $G(x^*) = 0$ (constraint active).

Substitute into $G(x^*) = 0$:

$$\begin{aligned} \sum_i [e_i * (a_i - \lambda * e_i / \alpha_i) - b_i] &= 0 \\ \lambda &= \sum_i (e_i * a_i - b_i) / \sum_i (e_i^2 / \alpha_i) \end{aligned}$$

Compute λ numerically, then $x_i^* = a_i - \lambda * e_i / \alpha_i$. Finally: $F(x^*) = \sum_i (1/2) * \alpha_i * (x_i^* - a_i)^2$.

IMPORTANT: Verify:

- $0 \leq x_i^* \leq 1$ for all i (no clipping needed)
- $G(x^*) = 0$ (constraint active)

3. $\lambda > 0$ (dual variable positive, confirming constraint is active)

Print these values.

3. Algorithm Constants — Derive from Paper

Before running, compute and print ALL of these. Cross-check each definition against the paper's notation.

Mirror map:

Use Euclidean: $\psi_i(x) = (1/2)x^2$, so $\mu_i = 1$ for all i , $\mu = \min_i \mu_i = 1$. Bregman distance: $V_i(u, x) = (1/2)(u - x)^2$. Mirror-prox step reduces to projected gradient descent:

$$x_{i\{k,t+1\}} = \text{clip}(x_{i\{k,t\}} - \gamma h_{i\{k,t\}}, 0, 1)$$

Lipschitz constants:

$$\begin{aligned} L_{\{f_i\}} &= \max_{x_i \in [0,1]} |\nabla f_i(x_i)| \\ &= \alpha_i * \max(a_i, 1 - a_i) \end{aligned}$$

$$L_{\{g_i\}} = e_i$$

$$L_G = \sqrt{\sum_i e_i^2} \quad [\text{Eq. (5)}]$$

Subgradient bounds (exact gradients, no noise):

$$M_i = \max(L_{\{f_i\}}, L_{\{g_i\}}) \quad (\text{worst-case norm across both modes})$$

$$M_{\text{bar}_i} = M_i \quad (\text{a.s. bound} = \text{second-moment bound, no noise})$$

$$M = \sqrt{\sum_i M_i^2}$$

$$M_{\text{bar}} = \sqrt{\sum_i M_{\text{bar}_i}^2} \quad (= M \text{ in the exact case})$$

Prox diameter:

$$\begin{aligned} D_x^2 &= \sum_i \sup_{x_i \in [0,1]} V_i(x_i^*, x_i) \\ &= \sum_i \max((1/2)(x_i^* - 0)^2, (1/2)(x_i^* - 1)^2) \end{aligned}$$

$$D_x = \sqrt{D_x^2}$$

4. Corollary 2 Parameter Choices

Use the **constant stepsize / constant tolerance** regime from Corollary 2.

Total inner updates:

$$N = K * T$$

Stepsize and tolerance:

$$\begin{aligned}\gamma &= D_x / (M * \sqrt{N}) \\ \eta &= 4 * M * D_x / (\delta * \sqrt{N})\end{aligned}$$

Use $\delta = 0.1$.

Condition on T — Eq. (13):

$$T < (\mu * M^2) / (4 * \delta * L_G * M_{\bar{}})$$

Since $\mu = 1$ and $M_{\bar{}} = M$:

$$T < M / (4 * \delta * L_G)$$

Compute the RHS. Choose $\bar{T}$ = largest integer strictly below this bound. **This should give $T \gg 6$ with the suggested parameters.**

Drift budget:

$$\begin{aligned}r_{\text{drift}} &= (2 * L_G * M_{\bar{}} / \mu) * T * \gamma \\ &= (2 * L_G * M * T * D_x) / (\mu * M * \sqrt{N}) \\ &= (2 * L_G * T * D_x) / \sqrt{N}\end{aligned}$$

Verify penalty-free condition (Corollary 1):

Check that $\eta \geq 2 * r_{\text{drift}}$. If this holds, $P \leq 0$ and convergence bound simplifies. This should hold by construction under Condition (13).

Print all of these values before running the algorithm.

5. Algorithm Implementation

Implement Algorithm 1 exactly as stated in Section IV. Follow line by line:

```
python

# ----- Setup -----
# Precompute: gamma, eta, r_drift, T, K, x_star, F_star
# Initialize: x[i] = a[i] for all i (starts at unconstrained optimum,
#           which violates the constraint — forces early correction rounds)

# ----- Main loop -----
for k in range(K):
    # Communication phase: uses  $x^{k,0}$  (state at START of round)
    s = [g_i(x[i]) for i in range(m)]    # each agent sends scalar
    G_tilde_k = sum(s)                  # server aggregates
    b_k = 1 if G_tilde_k <= eta - r_drift else 0 # gate decision

    for t in range(T):
        # *** LOG FIRST — this records  $x^{k,t}$  BEFORE the update ***
        log(x.copy(), G(x), F(x), b_k)

        # Then update
        for i in range(m):
            if b_k == 1:
                h = alpha[i] * (x[i] - a[i]) # grad f_i
            else:
                h = e[i]                    # grad g_i

            x[i] = clip(x[i] - gamma * h, 0, 1)
        # x now holds  $x^{k,t+1}$ 

    # Warm start: x carries over ( $x = x^{k,T} = x^{k+1,0}$ ) automatically

# ----- Output average -----
# Collect all iterates from feasible rounds (b_k == 1)
# x_bar = (1 / |feasible iterates|) * sum of those iterates
# (With constant gamma, the gamma weighting cancels in the average)
```

Logging — record at EVERY iterate (k, t):

Store the following in arrays for plotting:

- $x^{k,t}$ (the joint vector)

- $G(x^{k,t}) = \sum_i g_i(x_i^{k,t})$
 - $F(x^{k,t}) = \sum_i f_i(x_i^{k,t})$
 - b_k (gate for this round)
 - Running feasible-round average $\bar{x}_n$ and gap $F(\bar{x}_n) - F(x^*)$
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6. Graphs to Produce

Graph 1: Convergence — $F(\bar{x}_n) - F(x^*)$ vs N (log-log)

- **x-axis:** Cumulative inner updates n (running from 1 to $N = K \cdot T$), **log scale**.
- **y-axis:** $F(\bar{x}_n) - F(x^*)$ where $\bar{x}_n$ is the running average over feasible-round iterates seen so far, **log scale**.
- **Overlay:** Dashed line showing theoretical bound $3M \cdot D_x / (\delta \cdot \sqrt{n})$ from Eq. (14).
- **Expected behavior:** Empirical curve decays with slope ≈ -0.5 on log-log, sitting below the theoretical bound.
- **Style:** Publication-quality matplotlib. Font size ≥ 10 . LaTeX axis labels: $F(\bar{x}_n) - F(x^*)$ and N (total inner updates). Include a legend for the empirical curve and theoretical bound.

Graph 2: Feasibility — $G(x^{k,t})$ vs iteration

- **x-axis:** Global iteration index (flattened $(k, t) \rightarrow$ single counter).
- **y-axis:** $G(x^{k,t})$ (scalar aggregate constraint value).
- **Horizontal dashed line:** ϵ (budget tolerance).
- **Color-coding:** Use background shading or segment coloring to distinguish feasible rounds ($b_k = 1$, blue) from infeasible rounds ($b_k = 0$, red/orange).
- **Expected behavior: A sawtooth / oscillating pattern:**
 - During feasible (blue) rounds: G drifts upward as robots chase their objectives, but stays strictly below ϵ . This is the drift bound at work.
 - During infeasible (red) rounds: G drops as robots perform constraint correction.
 - Over time, oscillations dampen as iterates converge toward x^* on the constraint boundary.
- **Key visual check:** During ALL feasible (blue) segments, $G(x^{k,t}) \leq \epsilon$. This is Theorem 1(i).
- **Style:** Same publication quality. LaTeX labels: $G(x^{k,t})$ and iteration . Legend/annotation for feasible vs infeasible shading.

Optional Graph 3: Convergence at larger m

- Same as Graph 1, but with $m = 10$ or $m = 20$ agents.
 - Generate parameters randomly: $\alpha_i \sim \text{Uniform}[3, 8]$, $a_i \sim \text{Uniform}[0.7, 1.0]$, $e_i \sim \text{Uniform}[0.1, 0.5]$, b_i chosen so $G(a) > 0$ and x_i^* is interior to $[0, 1]$ for all i .
 - Shows the $O(1/\sqrt{N})$ rate holds at scale.
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7. Verification Checklist (do ALL before plotting)

Print and verify each of these. **If any check fails, stop and debug.**

- ☐ x^* values and $F(x^*)$ (exact via KKT)
- ☐ $G(x^*) = 0$ (constraint active) and $\lambda > 0$
- ☐ All constants: $L_{\{f_i\}}$, $L_{\{g_i\}}$, L_G , M_i , $M_{\bar{i}}$, M , $M_{\bar{}}$, D_x , μ
- ☐ T value and verification that $T < M / (4 * \delta * L_G)$ strictly
- ☐ γ , η , r_{drift} values
- ☐ Penalty-free: $\eta \geq 2 * r_{\text{drift}}$
- ☐ After running: $|B|$ (# feasible rounds) $\geq K/2$ (Lemma 5(iii))
- ☐ $G(x^{\{k,t\}}) \leq \eta$ for ALL $k \in B$, ALL t (Theorem 1(i))
- ☐ Final gap $F(\bar{x}) - F(x^*)$
- ☐ Log-log slope of convergence curve ≈ -0.5

Common failure modes:

1. T violating condition (13) $\rightarrow$ reduce T
 2. η too small relative to r_{drift} $\rightarrow$ increase K (more total rounds)
 3. Wrong gradient sign (should be $\alpha_i^*(x_i - a_i)$, not $a_i - x_i$)
 4. Missing projection / clip to $[0, 1]$
 5. Off-by-one: round starts at $t=0$, warm-start is $x^{\{k,T\}}$
 6. Averaging over ALL iterates instead of only feasible-round iterates
 7. **Logging AFTER update instead of BEFORE** — the paper's output average uses $x^{\{k,t\}}$ (pre-update state), not $x^{\{k,t+1\}}$. Always log first, then update.
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8. Run Parameters

$K = 500$ (outer rounds; increase to 2000 if log-log range is too narrow)

$T = \langle \text{computed from condition (13), should be } \gg 6 \rangle$

$\delta = 0.1$

$s = 0$ (start from round 0)

$m = 2$

Initialization: $x_i^{0,0} = a_i$ for all i

If the feasibility graph shows very few switches (all feasible or all infeasible), the problem may be too easy or too hard. Adjust $\overline{b}_i$ values to control how tight the budget is: smaller $\overline{b}_i \rightarrow$ tighter budget $\rightarrow$ more infeasible rounds $\rightarrow$ more visible switching.

9. Output Files

Save graphs as **high-resolution PDFs** (for LaTeX) and PNGs (for quick inspection):

- $\text{convergence_synthetic.pdf}$ / .png
- $\text{feasibility_synthetic.pdf}$ / .png

Save a text log of all computed constants and verification results:

- $\text{experiment_log.txt}$

Save the raw data arrays (pickle or .npz) so we can re-plot if needed:

- $\text{experiment_data.npz}$